Is there no way out? Is there no way to make a wormhole without getting entangled with the ill-understood laws of quantum gravity—no *perfectly classical* way?

Somewhat surprisingly, there *is*—but only if one pays a severe price. In 1966, Robert Geroch (a student of Wheeler's at Princeton) used global methods to show that one *can* construct a wormhole by a smooth, singularity-free warping and twisting of spacetime, but one can do so only if, during the construction, time also becomes twisted up as seell in all reference frames.[7] More specifically, while the construction is going on, it must be possible to travel backward in time, as well as forward; the "machinery" that does the construction, whatever it might be, must function briefly as a time machine that carries things from late moments of the construction back to early moments (but not back to moments before the construction began).

The universal reaction to Geroch's theorem, in 1967, was "*Surely* the laws of physics forbid time machines, and thereby they will prevent a wormhole from ever being constructed classically, that is, without tearing holes in space."

In the decades since 1967, some things we thought were *sure* have been proved wrong. (For example, we would never have believed in 1967 that a black hole can evaporate.) This has taught us caution. As part of our caution, and triggered by Sagan-type questions, we began asking in the late 1980s, "Do the laws of physics *really* forbid time machines, and if so, *how?* How might the laws enforce such a prohibition?" To this question I shall return below.

Let us now pause and take stock. In 1993 our best understanding of wormholes is this:

If no wormholes were made in the big bang, then an infinitely advanced civilization might try to construct one by two methods, quantum (pulling it out of the quantum foam) or classical (twisting spacetime without tearing it). We do *not* understand the laws of quantum gravity well enough to deduce, in 1993, whether the quantum construction of wormholes is possible. We *do* understand the laws of classical gravity (general relativity) well enough to know that the classical construction of wormholes is permitted only if the construction machinery, whatever it might be, twists